The map induced on the total space of the cover is 0-connected

The Univalent Foundations Program

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Editorial note. The counted statement is the author lem environment at source lines 2161-2163 and its proof environment opens on line 2164, the next line; the pairing is by adjacency and the retained proof range 2164-2177 is the whole environment.

Editorial note. The book label-number dump gives this label as 8.7.9 (env lem, chapter 8, section 7, ninth counter value), so the counters chapter 8, section 7 and thm 8 (zero-based) are set so that the item prints with the book number.

Editorial note. The counted statement and proof refer to no label outside the card: the retained ranges carry no cross reference command, and the connectedness and fibre facts used in the proof are recorded as standard prerequisites rather than treated as delegations.

Editorial note. The authors close the path-constructor case with the sentence that the equalities in question lie in a set and hence the case is automatic, quoted here verbatim; that is the argument itself, not an omission of a substantive step.

Editorial note. The retained preamble is the book own macro library and layout adapter as stored for this chapter; no package outside it is requested.

Editorial scope. One original result of the pinned HoTT Book source tree, counted once: the induced map on the total space of the cover is 0-connected (author label thm:kbar, book Lemma 8.7.9).

Allowed standard prerequisites (editorial).

- Connectedness of a map through its homotopy fibres. A map is 0-connected exactly when the 0-truncation of each of its homotopy fibres is contractible.. It is the unfolding of the goal at the start of the proof.
- The map k is surjective. The map from the set of generators to the base is (-1)-connected, that is, every element of the base is merely a value of it.. It lets the proof assume the point of the base is a value of the map, because contractibility is a mere proposition.
- Equality in a homotopy fibre and in a set. Two points of a homotopy fibre are equal as soon as the underlying points and the transported paths agree, and in a set all parallel equalities coincide.. The first gives the equality of the chosen point with an arbitrary point, and the second discharges the case of the path constructor automatically.

Lemma 8.7.9. *$\overline{k}$ is 0-connected.*

Proof. We must show that for all $a : A$, the 0-truncation of the type $\sum_{(t:T)}(\overline{k}t = a)$ is contractible. Since contractibility is a mere proposition and k is (-1)-connected, we may assume that $a = ks$ for some $s : S$. Now we can take the center of contraction to be $|(\ell s, q)|_0$ where q is the equality $\overline{k}\ell s = ks$.

It remains to show that for any $\phi : \left\|\sum_{(t:T)}(\overline{k}t = ks)\right\|_0$ we have $\phi = |(\ell s, q)|_0$. Since the latter is a mere proposition, and in particular a set, we may assume that $\phi = |(t, p)|_0$ for $t : T$ and $p : \overline{k}t = ks$.

Now we can do induction on $t : T$. If $t \equiv \ell s'$, then $ks' = \overline{k}\ell s' \stackrel{p}{=} ks$ yields via m an equality $\ell s = \ell s'$. Hence by definition of $\overline{k}$ and of equality in homotopy fibers, we obtain an equality $(ks', p) = (ks, q)$, and thus $|(ks', p)|_0 = |(ks, q)|_0$. Next we must show that as t varies along m these equalities agree. But they are equalities in a set (namely $\left\|\sum_{(t:T)}(\overline{k}t = ks)\right\|_0$), and hence this is automatic. □